

Asad Ikram

MSc Data Analytics (Distinction) · Chevening Scholar 2024/25 · Data engineering, LLM annotation and validation
Applying for: PhD in Secure and Privacy-Preserving Collaborative Language Models, Aalborg University Copenhagen (ref. 2026/987)
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PROFILE

I build Python data systems on AWS and LLM labelling pipelines, and I check machine labels against human ones before models are trained on them. Large-scale automated web extraction since 2019 has taught me how query-based extraction looks to a defender and is detected. I want to bring both to keeping data confidential and models intact when parties build language models together.

EDUCATION

Sep 2024 - Sep 2025 awarded 8 Dec 2025	MSc Data Analytics, Distinction. University of Portsmouth, United Kingdom. Chevening Scholar. Credit-weighted average 70.5/100 (Distinction threshold 70). Master's project 78/100. Supervisor: Dr Ella Haig. 180 UK credits, conventionally mapped to 90 ECTS. Modules included Intelligent Data and Text Analytics, Big Data and Machine Learning, Explainable AI, Data Management (80). Taught and assessed in English. Dissertation: <i>Cross-Platform Sentiment Analysis of Public Reaction to UK Economic Policies</i>. About 279,000 public posts from X/Twitter, Reddit, YouTube and Quora, collected through official APIs and licensed feeds under university ethics approval (Portsmouth TETHIC-2025-111094) and GDPR-compliant procedures. Fine-tuned a BERT classifier against BiLSTM and lexicon baselines (macro-F1 0.878, scored against lexicon-seeded labels); 200 posts hand-labelled by me, with agreement with VADER's labels of Cohen's $\kappa \approx 0.78$; temperature-scaling calibration, McNemar and bootstrap tests, LIME and integrated gradients.
Aug 2017 - Jan 2022 conferred 31 Jan 2022	BS Computer Science. National University of Computer and Emerging Sciences (FAST-NUCES), Pakistan (Karachi, then Chiniot-Faisalabad campus). 130 credit hours completed. CGPA 2.75 on a 0.00 to 4.00 scale. Courses included Probability and Statistics (B+), Linear Algebra, Calculus I and II, Numerical Methods, Discrete Structures, Design and Analysis of Algorithms, Theory of Automata (B+), Operating Systems (A-) and Applied Machine Learning (B).

RESEARCH

Ikram, A. and Haig, E. "Is the Platform Part of the Measurement? A Protocol and Simulation Study for Cross-Platform Equivalence of Machine-Labelled Policy Sentiment." **Manuscript, complete (v1, October 2026); not yet submitted or published.** Code: github.com/asadfix/platform-measurement, Zenodo DOI [10.5281/zenodo.23105129](https://doi.org/10.5281/zenodo.23105129). Treats each social platform as a survey mode and applies generalizability theory, measurement invariance and differential item functioning, and score linking to machine-labelled text. In a 500-replication simulation with known ground truth, an index that pools platform-level scores at face value reports the wrong sign of the aggregate on roughly a quarter of days, from measurement differences between platforms alone. *No peer-reviewed publications to date.*

EXPERIENCE

2025 - present part-time	Co-Founder and Chief Technology Officer, Artemis AI (formerly ViralData), United Kingdom (remote). Pre-launch social-media analytics start-up; pilots from December 2026. I lead the technology in a team of eight. • LLM labelling chain, checked against people. The team first hand-labelled about 500 posts and comments and checked Claude's and DeepSeek's labels against them. In the chain I designed, Claude labels about 6,000 posts with a written explanation of each, rules mined from those explanations become a prompt for DeepSeek, and DeepSeek labels about 12,000 more to train our encoder classifiers (XLM-R, DistilBERT, DeBERTa-v3, Detoxify bases) served on SageMaker. • Auditing the labels. Comparing the two tiers, I found 82.9% agreement on sentiment, 74.4% on emotion and 0.0% on toxicity, traced to mismatched label taxonomies, while the toxicity model reported 96.7% average confidence. A model-quality audit I ran in September 2026 found that reported accuracies measured agreement with the labelling model rather than with people; I am now rebuilding evaluation and release gates around a human-labelled golden set before launch. • Art-E, in progress: leading the build of a retrieval-based advisor that answers each client's questions from that client's own data, for December pilots. Security design: tenant isolation at the database, retrieval scoped to one client's records, prompt-injection red-teaming before release, an output check that rejects numbers absent from the client's data, and an external penetration test before launch. • Data platform: Meta Graph API ingestion into a layered warehouse on AWS (London), now about 11.6 million rows across 24 tables; shaped the Meta OAuth and compliance strategy and steered the platform through Meta's API metrics deprecation.
Dec 2024 - present part-time	Data Engineer (part-time), M+C Saatchi Fluency, London (remote). Data consultancy analysing online conversation. • Designed and operate a fleet of 64 Scrapy crawlers collecting car-owner forum discussion for a global automotive brand (AWS EC2, RDS Postgres, S3, Athena), self-deploying behind a CI gate with about 1,200 offline tests, plus a self-repairing LLM agent that diagnoses a broken source, escalates between Claude models by cause, regenerates the spider and tests it on live pages before shipping. • Securing that agent: third-party page text is fenced as untrusted data in prompts; secrets and cookies are refused and caught by a scanner; generated code is checked against forbidden patterns; every merge needs a single-use approval code bound to the test sample and expiring in ten minutes; the build service that

	colleagues' agents call sits behind OAuth with PKCE; per-person model budgets; adversarial reviews by dozens of refuter agents, mutation tests and negative controls. - Operate and repair a global e-commerce client's weekly social-analytics ETL (AWS Lambda, Fargate, Postgres, Tableau; master table 2.2 million rows by 184 columns): memory fixes, gated translation repairs and an 89-check data-quality sweep before each release. LLM-assisted text analysis, checked by hand: analyst on a study of twelve months of social discourse (100,845 Brandwatch and 170,542 Reddit mentions): BERTopic-derived codebook, LLM tagging with a prompt-version hash on every tag, and three pilot runs on the same 500 rows compared before the full run; contributor to LLM-assisted coding of about 170,000 multilingual posts and reviews for a global infant-nutrition brand.
Sep 2024 - present part-time, remote	Shorter engagements. CXG, Dubai, freelance data engineer in three engagements since Nov 2025: production crawlers and a pytest QA framework. VendueTech, Ireland, Lead Data Engineer, Jun to Jul 2025: led three engineers on real-time auction-data ingestion and an Azure-to-AWS migration (PySpark on EMR Serverless). Prefe, Milan, Data Engineer, Sep to Dec 2024: moved legacy collectors to Scrapy, cutting the failure rate by 60%; mentored two junior engineers.
May 2023 - present contract, part-time	Data Engineer (contract), Fix.com, Canada (remote). Online replacement-parts retailer. - Since June 2026 I have designed and built, alone, the company's crawler platform for replacement-parts catalogues: control-plane dashboard, Scrapy spiders, a data-quality gate and delivery (about 331 commits, all mine). By October 2026 it had delivered 10 million rows across 82 datasets. - Earlier: Scrapy/Scrapyd pipelines processing 50M+ data points from 200+ sources; Great Expectations validation with automated schema-drift detection, cutting bad production data by 70%.
Feb 2022 - Aug 2024 full-time, on-site	Associate Software Engineer, then Data Engineer, then Senior Data Engineer, Dubizzle Labs (Dubizzle, Bayut, Zameen, OLX), Lahore, Pakistan. Promoted twice in about two years. - Led a three-engineer team running the regional scraper fleet: 500+ scrapers across 15 markets on Kubernetes/EKS, collecting multilingual platform data in English, Arabic, Urdu and Hindi. - Redesigned the layered warehouse (RDL, ODL, ADL), raising ETL/ELT throughput by 60%. Built competitor-analysis dashboards from e-commerce and real-estate listings and online discussions that predicted revenue and showed what customers said.
Dec 2018 - Feb 2026 part-time	Freelance data engineer, Fiverr. 300+ completed orders in web scraping, automation, ETL and data analysis for international buyers; Level Two seller (first reached August 2021).

SKILLS

Programming	Python (my production language: Scrapy, FastAPI, PyTorch), PySpark, SQL; Git and GitHub Actions CI/CD; R (functional).
ML and NLP	Fine-tuning encoder classifiers (BERT, XLM-R, DistilBERT, DeBERTa-v3, Detoxify); LLM-assisted labelling through the Claude and DeepSeek APIs, including mining decision rules from model-written explanations; confidence-weighted loss and class-imbalance handling; BERTopic; CLIP, OCR, Whisper; BiLSTM.
Evaluation and validity	Checking model and LLM labels against hand-labelled sets; macro-F1; Cohen's κ ; temperature-scaling calibration; McNemar and bootstrap tests; LIME and integrated gradients; generalizability theory, measurement invariance and DIF, score linking.
Cloud and distributed	AWS (SageMaker, ECS Fargate, Lambda, Step Functions, Glue, Athena, Redshift, S3, RDS Postgres, DynamoDB, EMR Serverless, Batch, EC2); Docker, Kubernetes/EKS; Airflow; PostgreSQL.
Automated extraction and bot detection	Scrapy/Scrapyd, Playwright, Selenium; official platform APIs (Meta Graph, Reddit, YouTube) and licensed feeds. Working knowledge, from years on the client side, of how commercial bot-management systems (Akamai, Cloudflare, DataDome, PerimeterX, Kasada) tell automated clients from people: TLS and browser fingerprints, request rate, timing and repetition, and challenge pages.
Security and data protection in practice	Hardening LLM agents that read untrusted web content (prompt fencing, secret scanning, forbidden-pattern checks, single-use approval gates, OAuth with PKCE); ethics-approved, GDPR-compliant collection through official APIs and licensed feeds (MSc); OAuth-based platform access, tenant isolation and compliance planning (Artemis AI); data-quality and release gates with human sign-off.

AWARDS, PUBLIC WORK AND LANGUAGES

Chevening	Chevening Scholar 2024/25: the UK Government (FCDO) global scholarship, which funded the MSc.
Public work	Author and maintainer of web-scraping-guide.com (since May 2026), a free-to-read engineering guide to web data collection, covering how bot-management systems detect automated clients and the legal and ethical limits of collection (23 sections as of September 2026). Invited by Zyte's developer-relations team to review a pre-release version of its open-source agentic scraping toolkit (August 2026).
Languages	Urdu (native). English: working language with employers and clients in six countries; MSc taught and assessed in English in the UK (Distinction).

REFEREES

Dr Ella Haig, Associate Professor in Artificial Intelligence, School of Computing, University of Portsmouth. ella.haig@port.ac.uk. MSc dissertation supervisor and co-author.

Dr Fahad Ahmad, Senior Lecturer, Centre for Cybercrime and Economic Crime, University of Portsmouth. Fahad.Ahmad@port.ac.uk. Taught and assessed me on the MSc.